

Bioavailability of oxalic acid from spinach, sugar beet fibre and a solution of sodium oxalate consumed by female volunteers.

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Oxalate bioavailability from sugar beet fibre (40 g), spinach (25 g) and a solution of sodium oxalate (182 mg) was tested in nine women using a triplicated 3 x 3 Latin square arrangement. Each test substance provided 120 mg oxalic acid. Throughout the study the volunteers consumed a control diet and the test substances were administered at breakfast on specified days. After an initial 2-day control period, oxalate was administered in three test periods that consisted of one test day followed by one control day. Urine collected during 24-hr periods was analysed daily for oxalate. Oxalate excretion did not differ among the five control days and was not increased significantly following the ingestion of sugar beet fibre by the volunteers. Oxalate excretion was greater (P less than 0.0001) for the mean of the spinach and sodium oxalate solution diets than for the mean of the sugar beet fibre and control diets. Oxalate bioavailability from sugar beet fibre was 0.7% compared with bioavailabilities of 4.5 and 6.2% for spinach and oxalate solutions, respectively. The low bioavailability of oxalate from sugar beet fibre may be attributable to its high ratio of minerals (calcium and magnesium) to oxalate, its complex fibre matrix or the loss of the soluble oxalate during processing of sugar beets.